

Multiple Choice (10 marks)

1. The walls of bronchioles contain
 - (a) cartilage and respiratory epithelium.
 - (b) Only respiratory epithelium
 - (c) Only cartilage
 - (d) Respiratory epithelium and cartilage
 - (e) smooth muscle and respiratory epithelium.
2. The creatine kinase reaction is:
 - (a) constantly activated, regardless of ATP levels.
 - (b) more efficient than glycolysis.
 - (c) slow compared with glycolysis.
 - (d) not involved in energy production.
 - (e) irreversible.
3. Which of the following disorders does not show X-linked inheritance?
 - (a) Haemophilia A
 - (b) Fragile X syndrome
 - (c) Becker muscular dystrophy
 - (d) Color blindness
 - (e) Lesch-Nyhan syndrome
4. How are new polyomaviruses detailed
 - (a) Shot gun sequencing
 - (b) Cultivation in human neural cells
 - (c) Deep pyro sequencing (NGS)
 - (d) Monoclonal antibody techniques
5. Which products of ADP degradation increase in concentration in the blood during multiple sprint sports?
 - (a) Ammonia, urea and creatine.
 - (b) Ammonia, hypoxanthine and creatinine.
 - (c) Ammonia, creatine and uric acid.
 - (d) Urea, creatine and uric acid.
 - (e) Ammonia, urea and uric acid.

True / False (10 marks)

Answer True or False

6. True or False: Older adults generally perform very well on tests of semantic memory, which involves the recall of general knowledge and facts.
7. True or False: Biotin is primarily involved in the conversion of glucose to glycogen, facilitating energy storage in the liver.
8. True or False: A universal vaccine for all types of flu has been developed and is widely administered globally.
9. True or False: Measuring blood pressure in an elevated arm will accurately reflect the individual's true blood pressure levels.
10. True or False: First cousins share, on average, 1.0/4 of their genes due to their familial relationship.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of a relative risk ratio of 1.2 (95% CI: 1.1 to 1.8) for hip fractures among elderly women who exercise regularly. What conclusions can be drawn regarding exercise and fracture risk?
12. Discuss the role of dietary factors, specifically saturated fat, in influencing the risk of lung cancer, and analyze the underlying mechanisms that contribute to this relationship.

End of Paper